

A hardness theory for learned time steppers

Injected error of neural time-stepping methods of arbitrary order

Draft

Abstract

This note gives a theory for the injected error of a learned time stepper. A method is written as a known part plus a network slot. The hardness of a method is the Lipschitz constant of the flow map minus the known part. Under a capacity model, the injected error of a method is bounded by a power of its hardness. For Taylor methods of order p with exact lower slots, the hardness is of order h^p . For the direct step ($p = 0$), the hardness does not decrease with h . Errors in lower slots add to the hardness if they are functions of the state, and add to an irreducible floor otherwise. A test-mode calculus extends the results to unresolved modes, to methods that wrap the network in a Runge–Kutta integrator, and to methods that use past predicted states. The rollout error is the product of an amplification factor, analyzed in [1], and the injected error, analyzed here. Proofs are in Appendix A.

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1 Notation and terms

Table 1 lists the symbols. Table 2 lists the defined terms. Each term has one name and is used with that name throughout.

2 Setting

The system is $\dot{x} = f(x)$ with flow map Φ_t . The data are pairs $(x, \Phi_h(x))$ with x drawn from the data set $\mathcal{A}$.

Assumption 1 (Regularity). The data set $\mathcal{A}$ is compact and invariant, $\Phi_t(\mathcal{A}) = \mathcal{A}$ for all $t \in \mathbb{R}$. For every order k used below, $x^{(k)}$ is bounded and Lipschitz on $\mathcal{A}$. The derivatives are defined by $x^{(0)}(x) = x$, $x^{(1)} = f$, and $x^{(k+1)} = Dx^{(k)} f$.

For chaotic dissipative systems, $\mathcal{A}$ is the global attractor. Its intrinsic dimension d can be much smaller than D .

3 Methods and slots

Definition 1 (Method). A method $\mathcal{S}$ is a step map

$$x_{j+1} = G_{\mathcal{S}}(x_j; F_1, \dots, F_{p-1}; \mathcal{N}_{\theta}). \quad (1)$$

The estimates $F_k \approx x^{(k)}$, $k = 1, \dots, p-1$, fill the lower slots. The network $\mathcal{N}_{\theta}$ fills the network slot. The integer $p \geq 0$ is the order of the network slot.

Definition 2 (Slot types). A lower slot is an exact slot if $F_k = x^{(k)}$. It is a state-based slot if F_k is a function of the current state x_j only. It is a history-based slot if F_k depends on predicted states $x_{j-1}, x_{j-2}, \dots$.

A finite-difference estimate is state-based during training, because it uses true states. The same estimate is history-based during rollout, because it uses predicted states.

Table 1: Symbols.

Symbol	Meaning
f	vector field of $\dot{x} = f(x)$, $x \in \mathbb{R}^{n_x}$
g	$Df f$, the second time derivative as a function of the state
Φ_t	flow map of f over time t
h	time step
t_{end}	rollout horizon; $J = t_{\text{end}}/h$ is the number of rollout steps
$\mathcal{A}$	data set, a compact invariant set of Φ_t
d	intrinsic dimension of $\mathcal{A}$
$x^{(k)}$	k -th time derivative as a function of the state; $x^{(0)}(x) = x$
M_k, L_k	$\sup_{\mathcal{A}} \ x^{(k)}\ $ and $\text{Lip}_{\mathcal{A}}(x^{(k)})$; $L = L_1$
$\text{Lip}_{\mathcal{A}}(u)$	Lipschitz constant of u on $\mathcal{A}$
$\ u\ _{\infty}$	$\sup_{x \in \mathcal{A}} \ u(x)\ $
$\mathcal{S}$	a method
$G_{\mathcal{S}}$	step map of the method $\mathcal{S}$
p	order of the network slot
F_k	estimate in the lower slot of order k
$e_k, e_k^{\text{sys}}, e_k^{\text{ran}}$	slot error, its systematic part, its random part
$\varepsilon_{\text{sys}}^{(k)}, \varepsilon_{\text{ran}}^{(k)}$	$\text{Lip}_{\mathcal{A}}(e_k^{\text{sys}})$ and $\sup \ e_k^{\text{ran}}\ $
c_k	slot weight of the slot of order k
Q_h, Q_h^{ex}	known part, and known part with exact lower slots
$\mathcal{N}_{\theta}$	network
$\mathcal{N}^*$	target of the network slot
$\mathcal{H}(\mathcal{S})$	hardness of the method $\mathcal{S}$
$\delta(\mathcal{S})$	injected error of the method $\mathcal{S}$
$\mathcal{E}(B, \tau)$	error function (Lipschitz bound B of the fitted function, label noise τ)
W, m	number of network weights, number of effective training samples
ν	observation noise standard deviation
ρ, α	rate and exponent of the capacity model
δ_{opt}	optimization floor
λ, z	test-mode rate, $z = h\lambda$
μ, μ^*	per-mode network value, per-mode value of the target
$R_{\mathcal{S}}(\mu; z)$	method multiplier
$\psi(z)$	known multiplier
R	stability polynomial of a Runge–Kutta method
ζ	$h\mu^*$
s, r	number of stages and order of a Runge–Kutta method
$a_{il}, b_i, \beta, \bar{a}$	Runge–Kutta coefficients, $\beta = \sum_i b_i $, $\bar{a} = \max_i \sum_l a_{il} $
κ_i	Runge–Kutta stages
Λ	Lipschitz bound of the trained network
q, ϕ_i, C_q	order, coefficients, and constant of a finite-difference estimate
ν'	bound on the noise amplitude
ϑ	scale factor in the homogeneity condition
w, v	off-manifold variable $\dot{x} - f(x)$, velocity estimate
ϵ_j	rollout error after j steps
χ, Γ	Lipschitz constant of the step map, amplification factor
ξ	parasitic root
$\exp_n(z)$	truncated exponential $\sum_{k=0}^n z^k/k!$
γ	growth rate per unit time of the learned step map

Table 2: Defined terms.

Term	Where defined	Short meaning
method, step map	Definition 1	rule that maps x_j to x_{j+1}
lower slot, network slot	Definition 1	places where estimates enter
exact slot, state-based slot, history-based slot	Definition 2	types of lower slot
slot error, systematic part, random part	Definition 3	$e_k = e_k^{\text{sys}} + e_k^{\text{ran}}$
affine method, known part, slot weight	Definition 4	$G_S = Q_h + c_p \mathcal{N}_\theta$
Runge–Kutta wrapper, implicit Euler wrapper	Section 6	network inside integrator stages
exponential known part	Definition 9	known multiplier $e^{z_{\text{lin}}}$
Taylor method of order p	Definition 5	known part is a Taylor sum
injected error	Definition 6	one-step error on $\mathcal{A}$
error function, capacity model	Assumption 3	network error versus Lipschitz bound
hardness	Definitions 7, 8	Lipschitz constant of what the network supplies
target	Definition 8	$\mathcal{N}^*$
test mode, method multiplier, known multiplier	Definition 9	linear scalar analysis
representable, per-mode hardness	Definition 10	existence and size of μ^*
parasitic root	Definition 10	extra root from history-based slots
resolved mode, unresolved mode	Definition 9	$ z \leq 1, z > 1$
rollout error, amplification factor	Section 8	error after J steps

Definition 3 (Slot error). The slot error of the lower slot of order k is $e_k = F_k - x^{(k)}$. It is split as $e_k = e_k^{\text{sys}} + e_k^{\text{ran}}$. The systematic part e_k^{sys} is a deterministic function of the state. The random part e_k^{ran} has zero mean conditional on the state. The sizes are $\varepsilon_{\text{sys}}^{(k)} = \text{Lip}_{\mathcal{A}}(e_k^{\text{sys}})$ and $\varepsilon_{\text{ran}}^{(k)} = \sup \|e_k^{\text{ran}}\|$.

Definition 4 (Affine method). A method is affine if

$$G_S = Q_h + c_p \mathcal{N}_\theta, \quad Q_h = \sum_{k=0}^{p-1} c_k F_k, \quad (2)$$

with $F_0 = x^{(0)}$. The map Q_h is the known part. The scalars c_k are the slot weights. The known part with exact lower slots is $Q_h^{\text{ex}} = \sum_{k=0}^{p-1} c_k x^{(k)}$.

Definition 5 (Taylor method of order p). The Taylor method of order p is the affine method with $c_k = h^k/k!$ for $k = 0, \dots, p$. The Taylor method of order 0 is the direct step, $G_S = \mathcal{N}_\theta$. The Taylor method of order 1 is the Euler step, $G_S = x + h\mathcal{N}_\theta$.

Definition 6 (Injected error). The injected error of a method is $\delta(\mathcal{S}) = \|G_S - \Phi_h\|_\infty$.

4 Capacity model

Assumption 2 (Observation noise). The observed next state is $\Phi_h(x) + \eta$ with η independent of x , zero mean, and standard deviation ν . The input state is observed without noise.

Assumption 3 (Capacity model). Consider a network class with W weights and a final linear output layer. Train it on m effective samples to regress a function u with $\text{Lip}_{\mathcal{A}}(u) \leq B$ and label noise of standard deviation τ . The trained network $\hat{u}$ satisfies

$$\|\hat{u} - u\|_{\infty} \leq \mathcal{E}(B, \tau) + \delta_{\text{opt}}. \quad (3)$$

The error function $\mathcal{E}$ is nondecreasing in B and jointly homogeneous of degree one,

$$\mathcal{E}(\vartheta B, \vartheta \tau) = \vartheta \mathcal{E}(B, \tau) \quad \text{for all } \vartheta > 0. \quad (4)$$

At fixed noise level ν , the error function has the form

$$\mathcal{E}(B, \nu) = \rho(W, m, \nu) B^{\alpha}, \quad 0 < \alpha \leq 1. \quad (5)$$

Two regimes give standard values of ρ and α for Lipschitz functions on a set of dimension d . In the approximation-limited regime, $\alpha = 1$ and $\rho \propto W^{-1/d}$ up to logarithmic factors [4]. In the data-limited regime, $\alpha = d/(d+2)$ and $\rho \propto \nu^{2/(d+2)} m^{-1/(d+2)}$ up to logarithmic factors [3]. Both forms satisfy Eq. (4). For samples taken from one trajectory, m is the effective sample size after a mixing correction [2].

Lemma 1 (Output scaling does not change the injected error). *For an affine method with exact lower slots, under Assumptions 2 and 3,*

$$\delta(\mathcal{S}) \leq \mathcal{E}(\text{Lip}_{\mathcal{A}}(\Phi_h - Q_h^{\text{ex}}), \nu) + \delta_{\text{opt}} |c_p|. \quad (6)$$

The bound depends on c_p only through the optimization floor term.

Lemma 1 is proved in Appendix A.1. The injected error depends on what the known part leaves for the network. It does not depend on how the network output is scaled.

5 Hardness of affine methods

For an affine method with exact lower slots, the network is trained to fit $(\Phi_h - Q_h^{\text{ex}})/c_p$.

Definition 7 (Hardness of an affine method). The hardness of an affine method is

$$\mathcal{H}(\mathcal{S}) = \text{Lip}_{\mathcal{A}}(\Phi_h - Q_h^{\text{ex}}). \quad (7)$$

Theorem 1 (Hardness bounds the injected error). *Let $\mathcal{S}$ be an affine method whose lower slots are exact or state-based. Under Assumptions 2 and 3,*

$$\delta(\mathcal{S}) \leq \rho \left[\mathcal{H}(\mathcal{S}) + \sum_{k=1}^{p-1} |c_k| \varepsilon_{\text{sys}}^{(k)} \right]^{\alpha} + \sum_{k=1}^{p-1} |c_k| \varepsilon_{\text{ran}}^{(k)} + |c_p| \delta_{\text{opt}}, \quad (8)$$

where $\rho = \rho(W, m, \nu_{\text{eff}})$ and $\nu_{\text{eff}}^2 = \nu^2 + \sum_k c_k^2 \text{Var}(e_k^{\text{ran}})$.

Theorem 1 is proved in Appendix A.2. The systematic part of a slot error adds to the hardness. The random part of a slot error adds to a floor that no network can remove.

Lemma 2 (Taylor known part). *Let $Q_h^{(n)} = \sum_{k=0}^n \frac{h^k}{k!} x^{(k)}$ with $n \geq 0$. Under Assumption 1,*

$$\|\Phi_h - Q_h^{(n)}\|_{\infty} \leq \frac{M_{n+1} h^{n+1}}{(n+1)!}, \quad (9)$$

$$\text{Lip}_{\mathcal{A}}(\Phi_h - Q_h^{(n)}) \leq \frac{L_{n+1} e^{hL} h^{n+1}}{(n+1)!}. \quad (10)$$

Lemma 3 (Direct step). *Under Assumption 1, $\|\Phi_h(x) - \Phi_h(x')\| \geq e^{-hL}\|x - x'\|$ for all $x, x' \in \mathcal{A}$. The hardness of the direct step satisfies $\mathcal{H} \geq e^{-hL}$.*

Lemmas 2 and 3 are proved in Appendices A.3 and A.4.

Corollary 1 (Hardness of Taylor methods). *For the Taylor method of order p with exact lower slots,*

$$\mathcal{H} = \text{Lip}_{\mathcal{A}}(\Phi_h - Q_h^{(p-1)}) \leq \frac{L_p e^{hL} h^p}{p!}, \quad p \geq 1, \quad (11)$$

and $\mathcal{H} \geq e^{-hL}$ for $p = 0$.

Corollary 1 follows from Lemmas 2 and 3 with $n = p - 1$. The Euler step has hardness at most $L e^{hL} h$. The direct step has hardness at least e^{-hL} . This is the identity argument. The Taylor method of order p removes one more power of h for each exact lower slot.

Rule 1 (Order selection). The bound in Eq. (11) for order p is smaller than the bound for order $p - 1$ if and only if

$$h L_p < p L_{p-1}. \quad (12)$$

Rule 1 follows from the ratio of the two bounds in Eq. (11), which is $h L_p / (p L_{p-1})$. For $p = 1$ it reads $h L < 1$, with $L_0 = 1$.

Lemma 4 (Finite-difference slot error). *Let $F_1(x) = h^{-1} \sum_{i=0}^q \phi_i \Phi_{-ih}(x)$ with coefficients that satisfy*

$$\sum_{i=0}^q \phi_i \frac{(-i)^k}{k!} = \begin{cases} 1 & k = 1, \\ 0 & k \neq 1, \end{cases} \quad k = 0, \dots, q. \quad (13)$$

Under Assumption 1, the systematic part of the slot error satisfies

$$\varepsilon_{\text{sys}}^{(1)} = \text{Lip}_{\mathcal{A}}(F_1 - f) \leq C_q L_{q+1} e^{qhL} h^q, \quad C_q = \sum_{i=0}^q |\phi_i| \frac{i^{q+1}}{(q+1)!}. \quad (14)$$

Lemma 4 is proved in Appendix A.5. The two-point backward difference has $q = 1$, $\phi_0 = 1$, $\phi_1 = -1$, and $C_1 = 1/2$. The three-point backward difference has $q = 2$, $(\phi_0, \phi_1, \phi_2) = (3/2, -2, 1/2)$, and $C_2 = 1$.

Rule 2 (Acceptable lower slots). In a Taylor method of order p , a state-based lower slot of order k keeps the hardness of order h^p if and only if $\varepsilon_{\text{sys}}^{(k)} = O(h^{p-k})$. For a finite-difference estimate of $x^{(1)}$ of order q , this condition is $q \geq p - 1$.

Rule 2 follows from Eq. (8) with $c_k = h^k/k!$, and from Eq. (14), which gives $c_1 \varepsilon_{\text{sys}}^{(1)} = O(h^{q+1})$.

Corollary 2 (Noise floor of finite-difference slots). *Under Assumption 2, a finite-difference estimate from Lemma 4 evaluated on noisy states has random part with $\varepsilon_{\text{ran}}^{(1)} \leq \nu' \sum_i |\phi_i|/h$, where ν' bounds the noise amplitude. The floor term $c_1 \varepsilon_{\text{ran}}^{(1)} \leq \nu' \sum_i |\phi_i|$ does not decrease with h . The order- p term of the flow map is larger than this floor only if*

$$h > \left(\frac{p! \nu' \sum_i |\phi_i|}{M_p} \right)^{1/p}. \quad (15)$$

Corollary 2 gives a lower limit on h . Rule 1 gives an upper limit on h .

6 Non-affine methods

A Runge–Kutta wrapper places the network inside stages. The step map is

$$G_S[\mathcal{N}_\theta](x) = x + h \sum_{i=1}^s b_i \kappa_i, \quad \kappa_i = \mathcal{N}_\theta\left(x + h \sum_{l<i} a_{il} \kappa_l\right), \quad (16)$$

for an explicit method with s stages. A method of the form Eq. (16) is a Runge–Kutta wrapper. The implicit Euler wrapper is the method $x_{j+1} = x_j + h\mathcal{N}_\theta(x_{j+1})$. Set $\beta = \sum_i |b_i|$ and $\bar{a} = \max_i \sum_l |a_{il}|$.

Assumption 4 (Target of the network slot). There is a map $\mathcal{N}^*$ with $G_S[\mathcal{N}^*] = \Phi_h$ on $\mathcal{A}$. For a Runge–Kutta method of order r , $\mathcal{N}^* = f + O(h^r)$ in $\text{Lip}_{\mathcal{A}}$.

For analytic f and small h , the map $\mathcal{N}^*$ in Assumption 4 is the inverse modified vector field of [5]. The expansion fails when $h|\lambda| \gtrsim 1$ for an eigenvalue λ of Df . Section 7 treats that case.

Assumption 5 (Wrapper regression). Training $G_S[\mathcal{N}_\theta]$ on flow-map data gives a network whose distance to $\mathcal{N}^*$ obeys Assumption 3 with $u = \mathcal{N}^*$ and label noise ν/h .

Lemma 5 (Runge–Kutta sensitivity). Let $\mathcal{N}_1, \mathcal{N}_2$ be defined on a set that contains all stage points, and let $\mathcal{N}_1$ be Λ -Lipschitz there. For the explicit method in Eq. (16),

$$\|G_S[\mathcal{N}_1] - G_S[\mathcal{N}_2]\|_\infty \leq h\beta(1 + h\Lambda\bar{a})^{s-1} \|\mathcal{N}_1 - \mathcal{N}_2\|_\infty. \quad (17)$$

Definition 8 (Hardness of a non-affine method). The map $\mathcal{N}^*$ in Assumption 4 is the target. For an affine method with exact lower slots, the target is $(\Phi_h - Q_h^{\text{ex}})/c_p$. The hardness of the method in Eq. (16) is $\mathcal{H}(\mathcal{S}) = h \text{Lip}_{\mathcal{A}}(\mathcal{N}^*)$.

Theorem 2 (Hardness of Runge–Kutta wrappers). Under Assumptions 3, 4, and 5, with Λ a Lipschitz bound for the trained network,

$$\delta(\mathcal{S}) \leq \beta(1 + h\Lambda\bar{a})^{s-1} \left[\mathcal{E}(\mathcal{H}(\mathcal{S}), \nu) + h\delta_{\text{opt}} \right]. \quad (18)$$

For a Runge–Kutta method of order r , $\mathcal{H}(\mathcal{S}) = h(L + O(h^r))$.

Lemma 5 and Theorem 2 are proved in Appendices A.6 and A.7. For methods with $b_i \geq 0$, $\beta = 1$. At small h , every Runge–Kutta wrapper has the same leading hardness hL as the Euler step. The order r changes the target $\mathcal{N}^*$ at order h^r . It also changes the behavior of unresolved modes, as shown in Section 7.

7 Test-mode calculus

Theorems 1 and 2 use expansions in hL . This section applies each method to one linear mode. The analysis holds for any value of $h\lambda$.

Definition 9 (Test mode). The test mode is $\dot{x} = \lambda x$ with $\lambda \in \mathbb{C}$ and $z = h\lambda$. In the test mode, the network slot is $\mathcal{N}_\theta(x) = \mu x$ with $\mu \in \mathbb{C}$. The method multiplier $R_S(\mu; z)$ is defined by $G_S(x) = R_S(\mu; z)x$ when all history-based slots use the true trajectory. For an affine method, $R_S(\mu; z) = \psi(z) + c_p\mu$, where $\psi(z)$ is the known multiplier. A mode is a resolved mode if $|z| \leq 1$ and an unresolved mode if $|z| > 1$. An exponential known part is a known part with known multiplier $e^{z_{\text{lin}}}$, where $z_{\text{lin}} = h\lambda_{\text{lin}}$ and λ_{lin} is the rate of the exact linear operator on the mode.

Definition 10 (Representable, per-mode hardness, parasitic root). The test mode is representable by $\mathcal{S}$ if some μ^* satisfies $R_{\mathcal{S}}(\mu^*; z) = e^z$. If it is not representable, the method has the irreducible per-step error $\min_{\mu} |R_{\mathcal{S}}(\mu; z) - e^z|$. The per-mode hardness is

$$\mathcal{H}(\mathcal{S}; z) = |\partial_{\mu} R_{\mathcal{S}}(\mu^*; z) \mu^*|. \quad (19)$$

A parasitic root is a root of the characteristic polynomial of the rollout recurrence other than e^z . It exists only for methods with history-based slots.

For an affine method, $\mathcal{H}(\mathcal{S}; z) = |e^z - \psi(z)|$. This equals Definition 7 applied to the test mode, because $\Phi_h - Q_h^{\text{ex}}$ is then the map $x \mapsto (e^z - \psi(z))x$. For the method in Eq. (16), $\mathcal{H}(\mathcal{S}; z) = |\zeta R'(\zeta)|$ with $\zeta = h\mu^*$. The factor $|\zeta|$ equals Definition 8 applied to the test mode. The factor $|R'(\zeta)|$ is the exact test-mode value of the sensitivity bound in Eq. (17).

Proposition 1 (Per-mode hardness of standard methods). *Table 3 gives the per-mode hardness, the representability condition, and the parasitic roots.*

Table 3: Per-mode hardness. Here $\exp_n(z) = \sum_{k=0}^n z^k/k!$ and R is the stability polynomial of the Runge–Kutta method. The exponential known part uses the exact linear part z_{lin} of z .

Method	$\mathcal{H}(\mathcal{S}; z)$	Representable	Parasitic root
Direct step	$ e^z $	all z	none
Euler step	$ e^z - 1 $	all z	none
Implicit Euler wrapper	$ e^z e^z - 1 $	all z	none
Explicit Runge–Kutta wrapper	$ \zeta R'(\zeta) , R(\zeta) = e^z$	$e^z \in R(\mathbb{C})$	none
Taylor order p , exact lower slots	$ e^z - \exp_{p-1}(z) $	all z	none
Taylor order 2, backward difference, history-based	$ e^z + e^{-z} - 2 $	all z	e^{-z}
Exponential known part	$ e^{z_{\text{lin}}} e^{z - z_{\text{lin}}} - 1 $	all z	none

Proposition 1 is proved in Appendix A.8. For affine methods, the per-mode hardness is the distance between e^z and the known multiplier. The direct step has known multiplier 0. The Euler step has known multiplier 1. The Taylor method of order p has known multiplier $\exp_{p-1}(z)$.

Proposition 2 (Crossovers on the real axis). *For real z :*

- (a) *The Euler step has smaller per-mode hardness than the direct step if and only if $z > -\ln 2$.*
- (b) *The implicit Euler wrapper has smaller per-mode hardness than the Euler step if and only if $z < 0$.*
- (c) *The implicit Euler wrapper has smaller per-mode hardness than the direct step if and only if $z < \ln 2$.*
- (d) *The Taylor method of order 2 with exact lower slots has smaller per-mode hardness than the direct step if and only if $z > -1$.*
- (e) *The Taylor method of order 2 with exact lower slots has smaller per-mode hardness than the Euler step if and only if $z > z_*$, where $z_* \approx -1.5936$ solves $2e^z - 2 - z = 0$.*
- (f) *The Taylor method of order 2 with the history-based backward difference has smaller per-mode hardness than both the direct step and the Euler step if and only if $z > -\ln 2$.*

(g) As $z \rightarrow -\infty$, the Taylor method of order 2 with exact lower slots has per-mode hardness $|z| - 1 + e^z$, which grows without bound.

Proposition 3 (Imaginary axis). *For $z = i\omega$ with real ω , the per-mode hardness of the direct step is 1, of the Euler step is $2|\sin(\omega/2)|$, of the Taylor method of order 2 with exact lower slots is $|e^{i\omega} - 1 - i\omega| = \omega^2/2 + O(\omega^3)$, and of the Taylor method of order 2 with the history-based backward difference is $4\sin^2(\omega/2)$.*

Proposition 4 (Representability limit of explicit Runge–Kutta wrappers). *For real ζ , the stability polynomial of the classical second-order Runge–Kutta method has minimum $1/2$ at $\zeta = -1$. The stability polynomial of the classical fourth-order Runge–Kutta method has minimum 0.2704 at $\zeta \approx -1.5961$. A decaying real test mode with e^z below this minimum is not representable by a real μ . The limits are $z < -\ln 2 \approx -0.693$ for the second-order method and $z < -1.308$ for the fourth-order method.*

Propositions 2, 3, and 4 are proved in Appendix A.9. A nonlinear network can couple two real modes into a complex pair, and complex ζ reaches smaller $|R(\zeta)|$. This removes the representability limit at the cost of a phase error.

Proposition 5 (Second-order reformulation). *Let $g = Df f$. Every solution of $\dot{x} = f(x)$ is a solution of $\ddot{x} = g(x)$ with $\dot{x}(0) = f(x(0))$. For any solution of $\ddot{x} = g(x)$, the off-manifold variable $w = \dot{x} - f(x)$ satisfies*

$$\dot{w} = -Df(x)w. \quad (20)$$

Lemma 6 (Volume of the two-step recurrence). *For the recurrence $x_{j+1} = 2x_j - x_{j-1} + c_2 \mathcal{N}_\theta(x_j)$, the map $(x_j, x_{j-1}) \mapsto (x_{j+1}, x_j)$ has Jacobian determinant 1 for every $\mathcal{N}_\theta$ and every c_2 .*

Proposition 5 and Lemma 6 are proved in Appendix A.10. Methods that integrate $\ddot{x} = g$ without resetting $\dot{x}$ to an estimate of $f(x)$ at each step contain the dynamics of Eq. (20). In the test mode, $Df = \lambda$ and w grows like $e^{-\lambda t}$. For a normal constant Df , the growth rate of w is $-\min \text{Re spec}(Df)$.

8 Rollout error

The rollout error is $\epsilon_J = \|\hat{x}_J - x_J\|$, where $\hat{x}_J$ is the predicted state and x_J is the true state.

Proposition 6 (Rollout error of one-step methods). *Let the method have no history-based slots. Let $\|G_S(x) - G_S(x')\| \leq \chi \|x - x'\|$ on a set that contains the true and predicted states. Then*

$$\epsilon_J \leq \delta(S) \frac{\chi^J - 1}{\chi - 1}, \quad \chi \neq 1, \quad (21)$$

and $\epsilon_J \leq J \delta(S)$ for $\chi = 1$. With $\chi = e^{\gamma h}$, the amplification factor is $(e^{\gamma t_{\text{end}}} - 1)/(e^{\gamma h} - 1)$.

Proposition 6 is proved in Appendix A.11. It is Eq. (20) of [1] with the injected error written out. The amplification factor is analyzed in [1]. The injected error is bounded by Theorems 1 and 2.

For methods with history-based slots, the test-mode rollout recurrence has characteristic roots e^z and ξ . The amplification factor over J steps is then of order $\sum_{j < J} \max(|e^z|, |\xi|)^j$. For the Taylor method of order 2 with the history-based backward difference, $\xi = e^{-z}$. On a decaying mode, $|\xi| > 1$.

9 Procedure for a proposed method

The following steps evaluate a proposed method. Each step points to the result it uses.

1. Write the step map in the form of Eq. (1). Classify each lower slot as an exact slot, a state-based slot, or a history-based slot (Definition 2).
2. If a lower slot is history-based, compute the parasitic roots in the test mode (Definition 10). If $|\xi| > 1$ on modes present in the data, reject the method or reset the slot from a state-based estimate at each step. If the method integrates $\ddot{x} = g$ with a tracked velocity, apply Proposition 5.
3. Check representability on the resolved modes and the unresolved modes (Definition 10, Proposition 4).
4. Compute the hardness. Use Corollary 1 or Theorem 2 for the resolved modes. Use Table 3 for the unresolved modes.
5. Add the slot errors. Add $\sum_k |c_k| \varepsilon_{\text{sys}}^{(k)}$ to the hardness and $\sum_k |c_k| \varepsilon_{\text{ran}}^{(k)}$ to the floor (Theorem 1). Check Rule 2 and Eq. (15).
6. Choose the capacity regime, which sets ρ and α from W , m , ν , and d (Assumption 3).
7. Compare two methods $\mathcal{S}$ and $\mathcal{S}'$ in the same capacity regime by

$$\frac{\epsilon_J(\mathcal{S})}{\epsilon_J(\mathcal{S}')} \approx \frac{\Gamma(\mathcal{S})}{\Gamma(\mathcal{S}')} \left(\frac{\mathcal{H}(\mathcal{S})}{\mathcal{H}(\mathcal{S}')} \right)^\alpha, \quad (22)$$

where Γ is the amplification factor. Eq. (22) holds when the floor terms are small compared with the hardness terms. The rate ρ cancels in Eq. (22). The network size enters through α and through the size of the floor terms relative to $\rho \mathcal{H}^\alpha$.

10 Worked cases

10.1 Direct step and Euler step

By Corollary 1, the Euler step has hardness at most $Le^{hL}h$ and the direct step has hardness at least e^{-hL} . By Theorem 1 with $\alpha = 1$, the ratio of the hardness bounds is at most $Le^{2hL}h$. By Proposition 2(a), a decaying mode with $z < -\ln 2$ has smaller per-mode hardness under the direct step. The Euler step is preferred on resolved modes. The direct step is preferred on decaying modes with $z < -\ln 2$.

10.2 Runge–Kutta wrappers

By Theorem 2, a Runge–Kutta wrapper of any order has leading hardness hL , the same as the Euler step. The order changes the target at order h^r . By Proposition 4, the fourth-order wrapper cannot represent decaying real modes with $z < -1.308$. By Proposition 2(b), the implicit Euler wrapper has smaller per-mode hardness than the Euler step on every decaying mode.

10.3 Taylor method of order 2 with exact lower slot

By Corollary 1, the hardness is at most $L_2 e^{hL} h^2/2$. By Rule 1, this bound is smaller than the Euler bound if $hL_2 < 2L$. By Proposition 2(g), the per-mode hardness grows like $|z|$ on decaying unresolved modes. The exact lower slot contributes the term z to the known multiplier, and the network must cancel it. An exponential known part for the linear operator removes this growth (Table 3).

10.4 Taylor method of order 2 with backward-difference lower slot

During training the slot is state-based. By Lemma 4 with $q = 1$, $\varepsilon_{\text{sys}}^{(1)} \leq L_2 e^{hL} h/2$. By Theorem 1, the hardness term is at most $L_2 e^{hL} h^2$, twice the exact-slot bound. During rollout the slot is history-based. The parasitic root is e^{-z} (Table 3). For decaying modes, $|e^{-z}| > 1$. The method is not suitable for rollout on dissipative systems unless the slot is reset from a state-based estimate at each step.

10.5 Runge–Kutta–Nyström integration of the second-order term

A Runge–Kutta–Nyström method for $\ddot{x} = \mathcal{N}_\theta(x)$ with velocity estimate v_j has the form

$$x_{j+1} = x_j + hv_j + h^2 \sum_i \bar{b}_i \mathcal{N}_\theta(X_i), \quad X_i = x_j + \tilde{c}_i hv_j + h^2 \sum_{l < i} \bar{a}_{il} \mathcal{N}_\theta(X_l), \quad (23)$$

with $\sum_i \bar{b}_i = 1/2$. Here $\tilde{c}_i$ are the stage times of the method. By Proposition 5, the true trajectory solves $\ddot{x} = g(x)$ with $\dot{x} = f(x)$. The source of v_j sets the behavior.

- (a) $v_j = f(x_j)$, an exact slot. The method is a one-step method. In the test mode, the known multiplier contains $1 + z$, and $\mathcal{H}(\mathcal{S}; z) = |e^z - 1 - z| (1 + O(|z|))$ on resolved modes. The leading hardness equals that of the Taylor method of order 2 with exact lower slots. Under Assumption 4 applied to Eq. (23), the target differs from g by $O(h)$ or less, with the power set by the order conditions of the tableau. On decaying unresolved modes the per-mode hardness grows like $|z|$, as in Proposition 2(g).
- (b) v_j from a backward difference of predicted states, a history-based slot. The rollout recurrence has the parasitic root e^{-z} .
- (c) v_j tracked as a state variable, $v_{j+1} = v_j + h \sum_i b_i \mathcal{N}_\theta(X_i)$. The continuous system contains Eq. (20). Deviations from $v = f(x)$ grow on decaying modes for any integrator.
- (d) v_j from a state-based network for $x^{(1)}$ with slot error e_1 . The method is a one-step method. By Rule 2, the order-2 hardness is kept if $\varepsilon_{\text{sys}}^{(1)} = O(h)$. The random part adds $h \varepsilon_{\text{ran}}^{(1)}$ to the floor.

Configurations (a) and (d) are one-step methods. Relative to the Taylor method of order 2, they change the target, not the leading hardness.

11 Testable predictions

- P1. At fixed W and m , the injected error scales like h^0 for the direct step, h^α for the Euler step, and $h^{p\alpha}$ for the Taylor method of order p with exact lower slots, on resolved modes (Corollary 1, Theorem 1).

- P2. At small h , the Euler step and Runge–Kutta wrappers of any order have injected errors that agree to leading order (Theorem 2).
- P3. The injected error spectrum per wavenumber follows the per-mode hardness of Table 3 evaluated at $z_k = h\lambda_k$.
- P4. The Taylor method of order 2 with exact lower slots has large injected error on decaying unresolved modes. An exponential known part removes it (Proposition 2(g)).
- P5. The fourth-order Runge–Kutta wrapper has an injected error floor on real decaying modes with $z < -1.308$ (Proposition 4).
- P6. The Taylor method of order 2 with the history-based backward difference diverges on dissipative systems and does not diverge from the parasitic root on systems with purely imaginary test-mode rates (Table 3).
- P7. Replacing an exact lower slot by a finite-difference estimate of order q keeps the order- p scaling of P1 if and only if $q \geq p - 1$ (Rule 2).
- P8. In the approximation-limited regime, the ratio of injected errors of two methods does not depend on W (Eq. (22)).

12 Status of the results

Lemmas 1, 2, 3, 4, 5, and 6 and Propositions 1–5 and 6 are proved under the stated assumptions. Theorems 1 and 2 are proved given Assumption 3, and Theorem 2 also needs Assumptions 4 and 5. Assumption 3 is a model. Its form with a single exponent α can be tested by fitting the injected error against the hardness across methods at fixed W and m . The test-mode calculus of Section 7 is exact for linear normal dynamics. For nonlinear dynamics it is a leading-order model on each mode. A proof that joins the resolved-mode results of Section 5 with the unresolved-mode results of Section 7 is not given here. An inertial-manifold splitting is one candidate.

A Proofs

A.1 Proof of Lemma 1

Let $U = \Phi_h - Q_h^{\text{ex}}$. The step error is $G_S - \Phi_h = c_p \mathcal{N}_\theta - U$. The network is trained on labels $y = (\Phi_h(x) + \eta - Q_h^{\text{ex}}(x))/c_p$. These are the function U/c_p plus label noise of standard deviation $\nu/|c_p|$. By Eq. (3),

$$\|\mathcal{N}_\theta - U/c_p\|_\infty \leq \mathcal{E}(\text{Lip}_{\mathcal{A}}(U)/|c_p|, \nu/|c_p|) + \delta_{\text{opt}}.$$

Multiply by $|c_p|$ and apply Eq. (4) with $\vartheta = |c_p|$:

$$\|c_p \mathcal{N}_\theta - U\|_\infty \leq \mathcal{E}(\text{Lip}_{\mathcal{A}}(U), \nu) + |c_p| \delta_{\text{opt}}.$$

The left side is $\delta(\mathcal{S})$. □

A.2 Proof of Theorem 1

Write $Q_h = Q_h^{\text{ex}} + \sum_k c_k e_k^{\text{sys}} + \sum_k c_k e_k^{\text{ran}}$. Define $U = \Phi_h - Q_h^{\text{ex}} - \sum_k c_k e_k^{\text{sys}}$. The step error is

$$G_S - \Phi_h = (c_p \mathcal{N}_\theta - U) + \sum_k c_k e_k^{\text{ran}}.$$

The labels are $U + \eta - \sum_k c_k e_k^{\text{ran}}$, divided by c_p . The noise term $\eta - \sum_k c_k e_k^{\text{ran}}$ has zero mean given the state and variance ν_{eff}^2 . By the argument of Appendix A.1, $\|c_p \mathcal{N}_\theta - U\|_\infty \leq \mathcal{E}(\text{Lip}_\mathcal{A}(U), \nu_{\text{eff}}) + |c_p| \delta_{\text{opt}}$. The Lipschitz seminorm is subadditive and absolutely homogeneous, so

$$\text{Lip}_\mathcal{A}(U) \leq \text{Lip}_\mathcal{A}(\Phi_h - Q_h^{\text{ex}}) + \sum_k |c_k| \text{Lip}_\mathcal{A}(e_k^{\text{sys}}) = \mathcal{H}(\mathcal{S}) + \sum_k |c_k| \varepsilon_{\text{sys}}^{(k)}.$$

Since $\mathcal{E}$ is nondecreasing in its first argument, Eq. (5) gives the first term of Eq. (8). The triangle inequality gives $\|\sum_k c_k e_k^{\text{ran}}\|_\infty \leq \sum_k |c_k| \varepsilon_{\text{ran}}^{(k)}$. $\square$

A.3 Proof of Lemma 2

Fix $x \in \mathcal{A}$. By invariance, $\Phi_t(x) \in \mathcal{A}$ for all t . By Assumption 1, $\frac{d^k}{dt^k} \Phi_t(x) = x^{(k)}(\Phi_t(x))$ for $k \leq n+1$. Taylor's theorem with integral remainder gives

$$\Phi_h(x) - Q_h^{(n)}(x) = \int_0^h \frac{(h-t)^n}{n!} x^{(n+1)}(\Phi_t(x)) dt. \quad (24)$$

The norm of the integrand is at most $M_{n+1}(h-t)^n/n!$. Integration gives $M_{n+1}h^{n+1}/(n+1)!$, which is Eq. (9).

For $x, x' \in \mathcal{A}$, subtract Eq. (24) at x' from Eq. (24) at x . The integrand difference is bounded by $\frac{(h-t)^n}{n!} L_{n+1} \|\Phi_t(x) - \Phi_t(x')\|$. By Grönwall's inequality on $\mathcal{A}$, $\|\Phi_t(x) - \Phi_t(x')\| \leq e^{tL} \|x - x'\| \leq e^{hL} \|x - x'\|$ for $0 \leq t \leq h$. Integration gives Eq. (10). $\square$

A.4 Proof of Lemma 3

By invariance, Φ_{-h} maps $\mathcal{A}$ to $\mathcal{A}$. By Grönwall's inequality applied backward in time, $\|\Phi_{-h}(\tilde{x}) - \Phi_{-h}(\tilde{x}')\| \leq e^{hL} \|\tilde{x} - \tilde{x}'\|$ for $\tilde{x}, \tilde{x}' \in \mathcal{A}$. Set $\tilde{x} = \Phi_h(x)$ and $\tilde{x}' = \Phi_h(x')$. Then $\|x - x'\| \leq e^{hL} \|\Phi_h(x) - \Phi_h(x')\|$. Rearranging gives the first claim. Taking $x \neq x'$ gives $\text{Lip}_\mathcal{A}(\Phi_h) \geq e^{-hL}$. The direct step has $Q_h^{\text{ex}} = 0$, so $\mathcal{H} = \text{Lip}_\mathcal{A}(\Phi_h)$. $\square$

A.5 Proof of Lemma 4

For $i \geq 0$, apply Taylor's theorem to $t \mapsto \Phi_t(x)$ on the interval from 0 to $-ih$:

$$\Phi_{-ih}(x) = \sum_{k=0}^q \frac{(-ih)^k}{k!} x^{(k)}(x) + T_i(x), \quad T_i(x) = \int_0^{-ih} \frac{(-ih-t)^q}{q!} x^{(q+1)}(\Phi_t(x)) dt.$$

Multiply by ϕ_i/h and sum over i . By Eq. (13), the coefficient of $h^{k-1}x^{(k)}$ is 1 for $k=1$ and 0 for $k \neq 1$. Therefore $F_1 - f = h^{-1} \sum_i \phi_i T_i$. For $x, x' \in \mathcal{A}$, the argument of Appendix A.3 run backward in time gives

$$\|T_i(x) - T_i(x')\| \leq L_{q+1} e^{ihL} \frac{(ih)^{q+1}}{(q+1)!} \|x - x'\|.$$

Sum over i with weights $|\phi_i|/h$ and use $e^{ihL} \leq e^{qhL}$ for $i \leq q$. This gives Eq. (14). For $q=1$, $(\phi_0, \phi_1) = (1, -1)$ satisfies Eq. (13), and $C_1 = 1 \cdot 1^2/2 = 1/2$. For $q=2$, $(\phi_0, \phi_1, \phi_2) = (3/2, -2, 1/2)$ satisfies Eq. (13), and $C_2 = (2 \cdot 1 + \frac{1}{2} \cdot 8)/6 = 1$. $\square$

A.6 Proof of Lemma 5

Let $\kappa_i^{(1)}$ and $\kappa_i^{(2)}$ be the stages for $\mathcal{N}_1$ and $\mathcal{N}_2$, with stage points $X_i^{(1)}$ and $X_i^{(2)}$. Set $\Delta_i = \|\kappa_i^{(1)} - \kappa_i^{(2)}\|$ and $\Delta_0 = \|\mathcal{N}_1 - \mathcal{N}_2\|_\infty$. Then

$$\Delta_i \leq \|\mathcal{N}_1(X_i^{(1)}) - \mathcal{N}_1(X_i^{(2)})\| + \|\mathcal{N}_1(X_i^{(2)}) - \mathcal{N}_2(X_i^{(2)})\| \leq \Lambda h \sum_{l < i} |a_{il}| \Delta_l + \Delta_0.$$

By induction, $\Delta_i \leq \Delta_0(1 + h\Lambda\bar{a})^{i-1}$. The case $i = 1$ holds because the sum is empty. If the bound holds for all $l < i$, then $\Delta_i \leq \Delta_0 + h\Lambda\bar{a} \Delta_0(1 + h\Lambda\bar{a})^{i-2} \leq \Delta_0(1 + h\Lambda\bar{a})^{i-1}$. The step maps differ by $h \sum_i b_i(\kappa_i^{(1)} - \kappa_i^{(2)})$, with norm at most $h\beta\Delta_0(1 + h\Lambda\bar{a})^{s-1}$. $\square$

A.7 Proof of Theorem 2

By Assumption 4, $G_S[\mathcal{N}_\theta] - \Phi_h = G_S[\mathcal{N}_\theta] - G_S[\mathcal{N}^*]$ on $\mathcal{A}$. By Lemma 5 with $\mathcal{N}_1 = \mathcal{N}_\theta$ and $\mathcal{N}_2 = \mathcal{N}^*$,

$$\delta(\mathcal{S}) \leq h\beta(1 + h\Lambda\bar{a})^{s-1} \|\mathcal{N}_\theta - \mathcal{N}^*\|_\infty.$$

By Assumption 5, $\|\mathcal{N}_\theta - \mathcal{N}^*\|_\infty \leq \mathcal{E}(\text{Lip}_{\mathcal{A}}(\mathcal{N}^*), \nu/h) + \delta_{\text{opt}}$. By Eq. (4) with $\vartheta = h$,

$$h \mathcal{E}(\text{Lip}_{\mathcal{A}}(\mathcal{N}^*), \nu/h) = \mathcal{E}(h \text{Lip}_{\mathcal{A}}(\mathcal{N}^*), \nu) = \mathcal{E}(\mathcal{H}(\mathcal{S}), \nu).$$

This gives Eq. (18). By Assumption 4, $\text{Lip}_{\mathcal{A}}(\mathcal{N}^*) = L + O(h^r)$. $\square$

A.8 Proof of Proposition 1

The exact multiplier of the test mode over one step is e^z .

- (i) Direct step. $R = \mu$, $\mu^* = e^z$, $\partial_\mu R = 1$, so $\mathcal{H} = |e^z|$.
- (ii) Euler step. $R = 1 + h\mu$, $h\mu^* = e^z - 1$, $\partial_\mu R = h$, so $\mathcal{H} = |e^z - 1|$.
- (iii) Implicit Euler wrapper, $x_{j+1} = x_j + h\mu x_{j+1}$. $R = 1/(1 - h\mu)$. The condition $R = e^z$ gives $h\mu^* = 1 - e^{-z}$, which exists for all z . $\partial_\mu R = h/(1 - h\mu)^2 = he^{2z}$ at μ^* . So $\mathcal{H} = |1 - e^{-z}| |e^{2z}| = |e^z| |e^z - 1|$.
- (iv) Explicit Runge–Kutta wrapper. $R_S = R(h\mu)$ with R the stability polynomial. Representability requires $e^z \in R(\mathbb{C})$. With $\zeta = h\mu^*$, $\partial_\mu R_S = hR'(\zeta)$, so $\mathcal{H} = |\zeta R'(\zeta)|$.
- (v) Taylor method of order p with exact lower slots. $x^{(k)}(x) = \lambda^k x$, so $\psi(z) = \sum_{k < p} z^k/k! = \exp_{p-1}(z)$. $\mathcal{H} = |e^z - \exp_{p-1}(z)|$.
- (vi) Taylor method of order 2 with backward difference. On the true trajectory, $x_{j-1} = e^{-z}x_j$, so $hF_1 = (1 - e^{-z})x_j$ and $\psi(z) = 2 - e^{-z}$. $\mathcal{H} = |e^z - 2 + e^{-z}|$. In rollout, $hF_1 = x_j - x_{j-1}$ and the network term equals $(e^z - 2 + e^{-z})x_j$ at μ^* . The recurrence is $x_{j+1} = (e^z + e^{-z})x_j - x_{j-1}$. Its characteristic polynomial is $\xi^2 - (e^z + e^{-z})\xi + 1 = (\xi - e^z)(\xi - e^{-z})$. The parasitic root is e^{-z} .
- (vii) Exponential known part. $\psi(z) = e^{z_{\text{lin}}}$, so $\mathcal{H} = |e^z - e^{z_{\text{lin}}}| = |e^{z_{\text{lin}}}| |e^{z - z_{\text{lin}}} - 1|$. $\square$

A.9 Proofs of Propositions 2, 3, and 4

All z are real in Proposition 2.

- (a) For $z \geq 0$, $e^z - 1 < e^z$. For $z < 0$, $1 - e^z < e^z$ if and only if $e^z > 1/2$, that is $z > -\ln 2$.
- (b) The ratio of the implicit Euler and Euler per-mode hardness is e^z , which is below 1 if and only if $z < 0$.
- (c) The ratio of the implicit Euler and direct-step per-mode hardness is $|e^z - 1|$, which is below 1 if and only if $z < \ln 2$.
- (d) $e^z - 1 - z \geq 0$ for all real z . The condition $e^z - 1 - z < e^z$ is $z > -1$.
- (e) For $z \geq 0$, $e^z - 1 - z < e^z - 1$. For $z < 0$, the condition is $e^z - 1 - z < 1 - e^z$, that is $2e^z - 2 - z < 0$. The function $2e^z - 2 - z$ is convex, is zero at $z = 0$ and at z_* , and is negative between them. Numerical root finding gives $z_* = -1.5936$.
- (f) Set $u = e^z$. The per-mode hardness is $u + 1/u - 2 \geq 0$. Against the direct step, $u + 1/u - 2 < u$ if and only if $u > 1/2$. Against the Euler step with $u < 1$, $u + 1/u - 2 < 1 - u$ is $2u^2 - 3u + 1 < 0$, that is $1/2 < u < 1$. Against the Euler step with $u \geq 1$, $u + 1/u - 2 < u - 1$ is $1/u < 1$, which holds for $u > 1$. Both conditions reduce to $z > -\ln 2$.
- (g) For $z < 0$, $|e^z - 1 - z| = e^z - 1 - z = |z| - 1 + e^z$.

For Proposition 3, $|e^{i\omega}| = 1$ and $|e^{i\omega} - 1| = 2|\sin(\omega/2)|$. The expansion $e^{i\omega} - 1 - i\omega = -\omega^2/2 + O(\omega^3)$ gives the order-2 value. Also $e^{i\omega} + e^{-i\omega} - 2 = 2\cos\omega - 2 = -4\sin^2(\omega/2)$.

For Proposition 4, the second-order polynomial $1 + \zeta + \zeta^2/2$ has derivative $1 + \zeta$, zero at $\zeta = -1$, with value $1/2$. The fourth-order polynomial $\exp_4(\zeta)$ has derivative $\exp_3(\zeta)$. This cubic has one real root, $\zeta \approx -1.5961$, found numerically. The value there is 0.2704 . Both polynomials tend to $+\infty$ as $\zeta \rightarrow \pm\infty$, so these values are the minima on the real line. The limits on z are $\ln(1/2) = -0.693$ and $\ln(0.2704) = -1.308$. $\square$

A.10 Proofs of Proposition 5 and Lemma 6

If $\dot{x} = f(x)$, then $\ddot{x} = Df(x)\dot{x} = Df(x)f(x) = g(x)$ and $\dot{x}(0) = f(x(0))$. For any solution of $\ddot{x} = g(x)$, set $w = \dot{x} - f(x)$. Then $\dot{w} = \ddot{x} - Df(x)\dot{x} = Df(x)f(x) - Df(x)\dot{x} = -Df(x)w$.

For Lemma 6, the Jacobian of $(x_j, x_{j-1}) \mapsto (x_{j+1}, x_j)$ is the block matrix

$$\begin{pmatrix} 2I + c_2 D\mathcal{N}_\theta(x_j) & -I \\ I & 0 \end{pmatrix}.$$

For a block matrix $\begin{pmatrix} Y_{11} & Y_{12} \\ Y_{21} & 0 \end{pmatrix}$ with $n_x \times n_x$ blocks, exchanging the two block columns gives the determinant $(-1)^{n_x} \det(Y_{12}) \det(Y_{21})$. Here $\det(Y_{12}) = (-1)^{n_x}$ and $\det(Y_{21}) = 1$, so the determinant is 1. $\square$

A.11 Proof of Proposition 6

By the triangle inequality,

$$\epsilon_{j+1} = \|G_S(\hat{x}_j) - \Phi_h(x_j)\| \leq \|G_S(\hat{x}_j) - G_S(x_j)\| + \|G_S(x_j) - \Phi_h(x_j)\|.$$

The first term is at most $\chi \epsilon_j$. The second term is at most $\delta(\mathcal{S})$ because $x_j \in \mathcal{A}$. With $\epsilon_0 = 0$, induction gives $\epsilon_J \leq \delta(\mathcal{S}) \sum_{j=0}^{J-1} \chi^j$. The geometric sum gives Eq. (21). $\square$

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